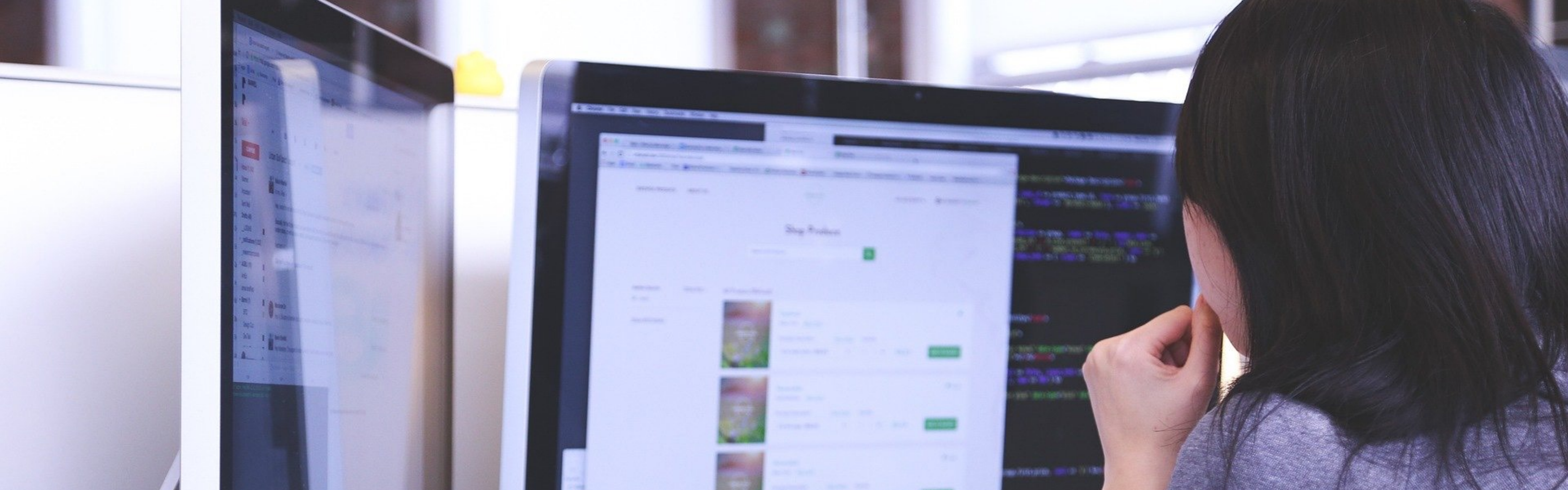
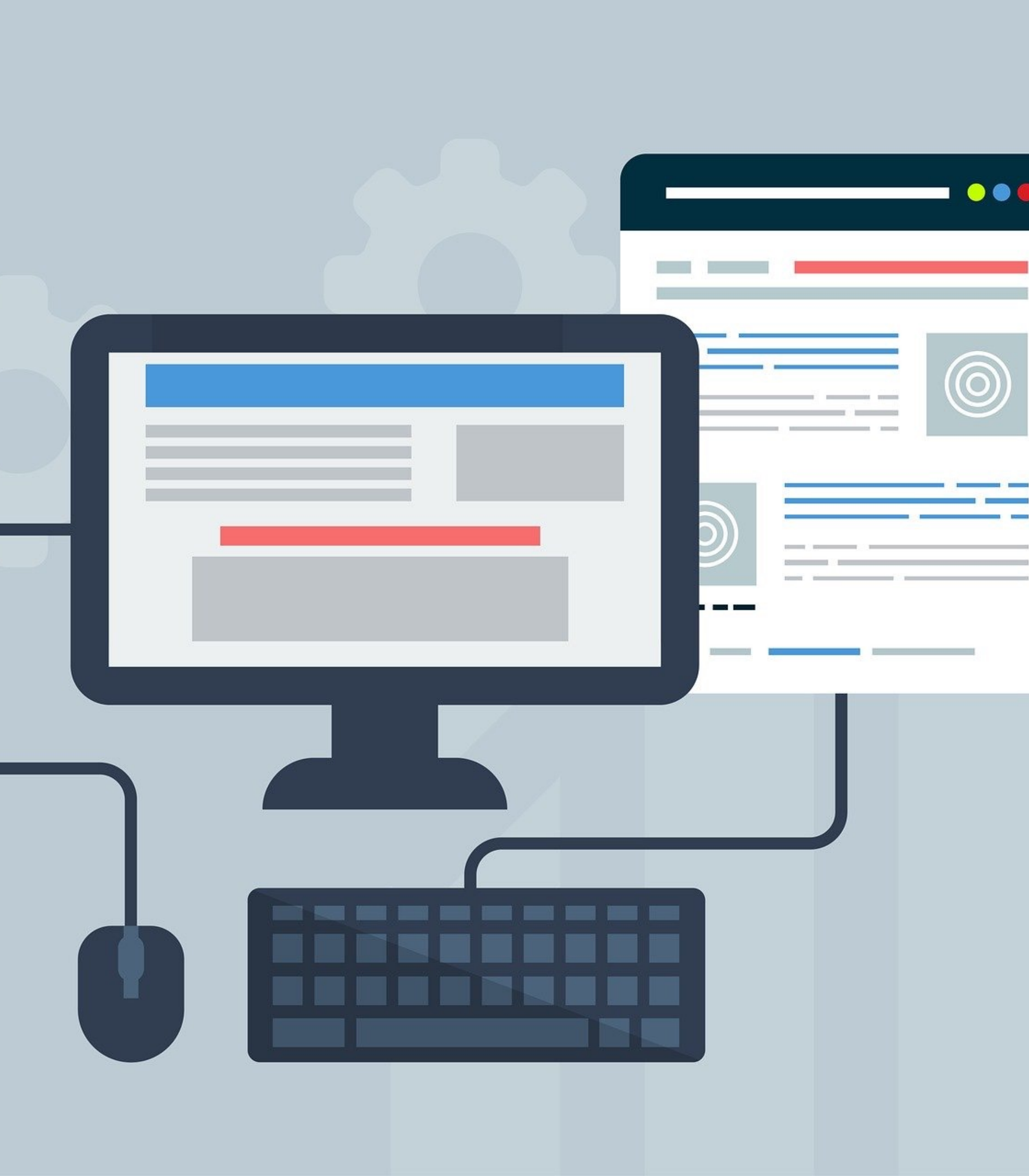




ARIA

Web Design Accessibility





INTRODUCTION TO ARIA

- Web Accessibility Initiative – Accessible Rich Internet Applications (WAI-ARIA)
- Created by W3C
- Part of HTML
- Allows you to make your website comprehensive to screenreaders, even if it is an interactive web application



USE OF ARIA

- ARIA is not designed to replace semantic HTML - this should always be your first port of call
- However, there are times you want to use ARIA instead of vanilla HTML
- For example, disable buttons with ARIA rather than the `disabled` attribute

ROLES

- The role attribute can be used to mimic a HTML element
- Again, we want to use semantic HTML first, and this as a last resort
- `role="navigation"`
- `role="checkbox"`
- `role="text"` (not part of specification)
- Avoid redundancy
 - `<nav role="navigation">`

ROLES (W3C EXAMPLE)

```
<div class="tabs">
  <div role="tablist" aria-label="Entertainment">
    <button role="tab" aria-selected="true" aria-controls="nils-tab" id="nils">Nils
Frahm</button>
    <button role="tab" aria-selected="false" aria-controls="agnes-tab" id="agnes"
tabindex="-1">Agnes Obel</button>
    <button role="tab" aria-selected="false" aria-controls="complexcomplex" id="complex"
tabindex="-1" data-deletable>Joke</button>
  </div>
  <div tabindex="0" role="tabpanel" id="nils-tab" aria-labelledby="nils">
    <p>Nils Frahm is a German musician.</p>
  </div>
  <div tabindex="0" role="tabpanel" id="agnes-tab" aria-labelledby="agnes" hidden>
    <p>Agnes Caroline Thaarup Obel is a Danish singer/songwriter.</p>
  </div>
  <div tabindex="0" role="tabpanel" id="complexcomplex" aria-labelledby="complex" hidden>
    <p>Fear of complicated buildings:</p>
    <p>A complex complex complex.</p>
  </div>
</div>
</div>
```

LABELS

- Labels can be added to anything and override other label attributes

```
<button class="menu-button" aria-label="Menu"></button>
```

HIDDEN

- Remove anything from the DOM that is purely presentational
- Do not hide anything important

```
<article>
```

```

```

```
<h2><a href="/news/clowns">Clowns demand higher pay</a></h2>
```

```
</article>
```

FORMS

- Labelled by establishes a link between an element and a label
- This is perfect for radios or checkboxes, for example
 - `<span aria-checked="true">`
- When “greying out” buttons you can use ARIA
 - `<button aria-disabled="true">Submit</button>`
- Add labels and additional notes (read after the form field) with ARIA
 - We’ll see an example on the next slide

FORMS

```
<span class="checkbox" aria-labelledby="newsletter" aria-  
describedby="newsletter-notes" aria-checked="false">
```

```
<label id="newsletter">Subscribe to our newsletter</label>
```

```
<span id="newsletter-notes">We send out one email per week</  
span>
```

LIVE REGIONS

- For parts of the page with dynamic updates, you can mark them as live regions
- This tells screen readers to read out any changes
- `<div aria-live="polite">...</div>`
- `<div aria-live="assertive">...</div>`
- For alerts and errors, there is already a live region role you can use
- `<div role="alert">Error message.</div>`